

Topics of today's class

Last class:

- Chapter 4
 - Behavior of the representative consumer

Today's class:

- Chapter 4
 - Behavior of the representative firm

The Representative Firm

- The production function.
- Profit maximization and labor demand.

How much does the economy produce?

What determines the quantity of goods and services produced? 2 Keys Factors of Production:

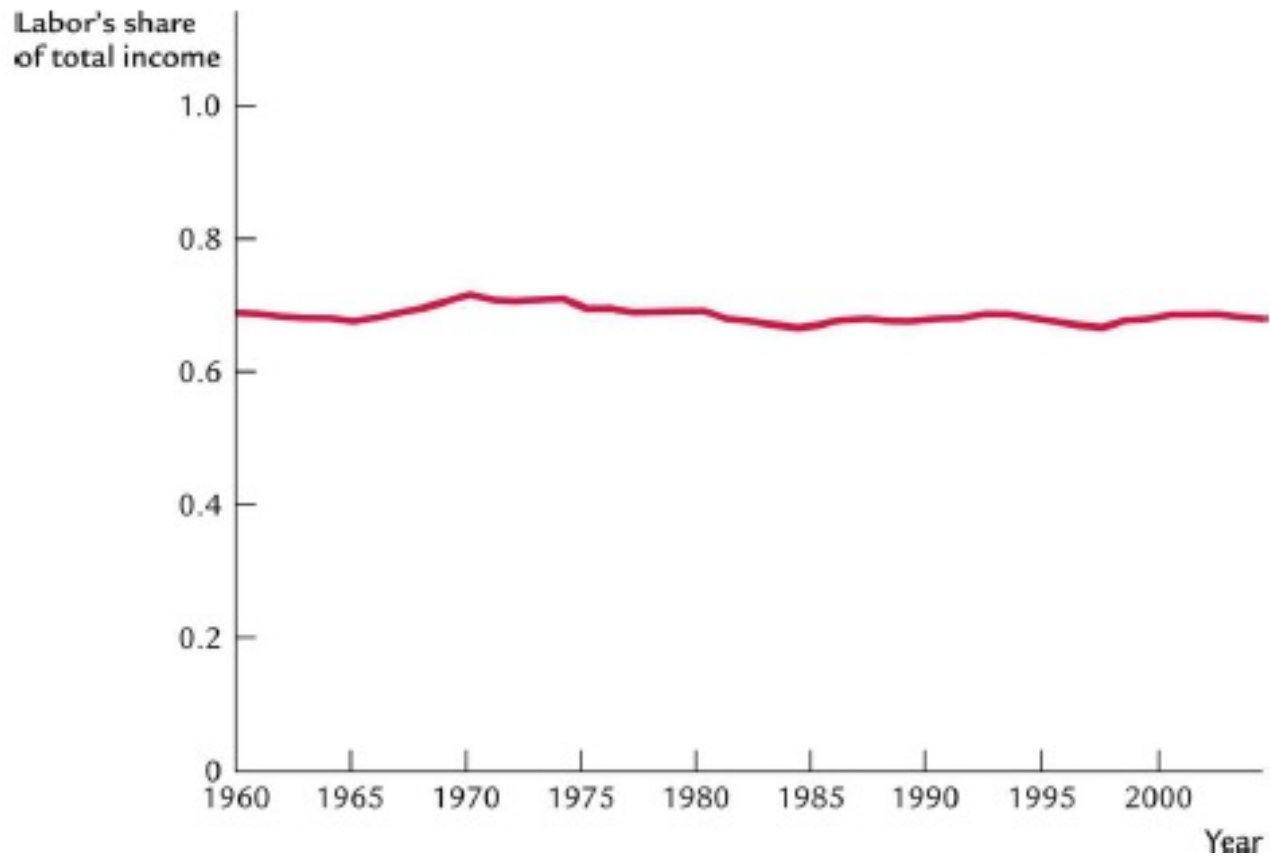
- K : Capital (factories and machines)
- N^d : Labor

Quality matters too:

- z is the level of technology or *total factor productivity* (TFP)
- Example: changes in the organization of production, personal computer, weather (especially in agriculture), government regulations, price of energy

$$Y = zF(K, N^d),$$

US aggregate production function



US aggregate production function

Empirical Studies show that a reasonable description of the U.S Economy production is:

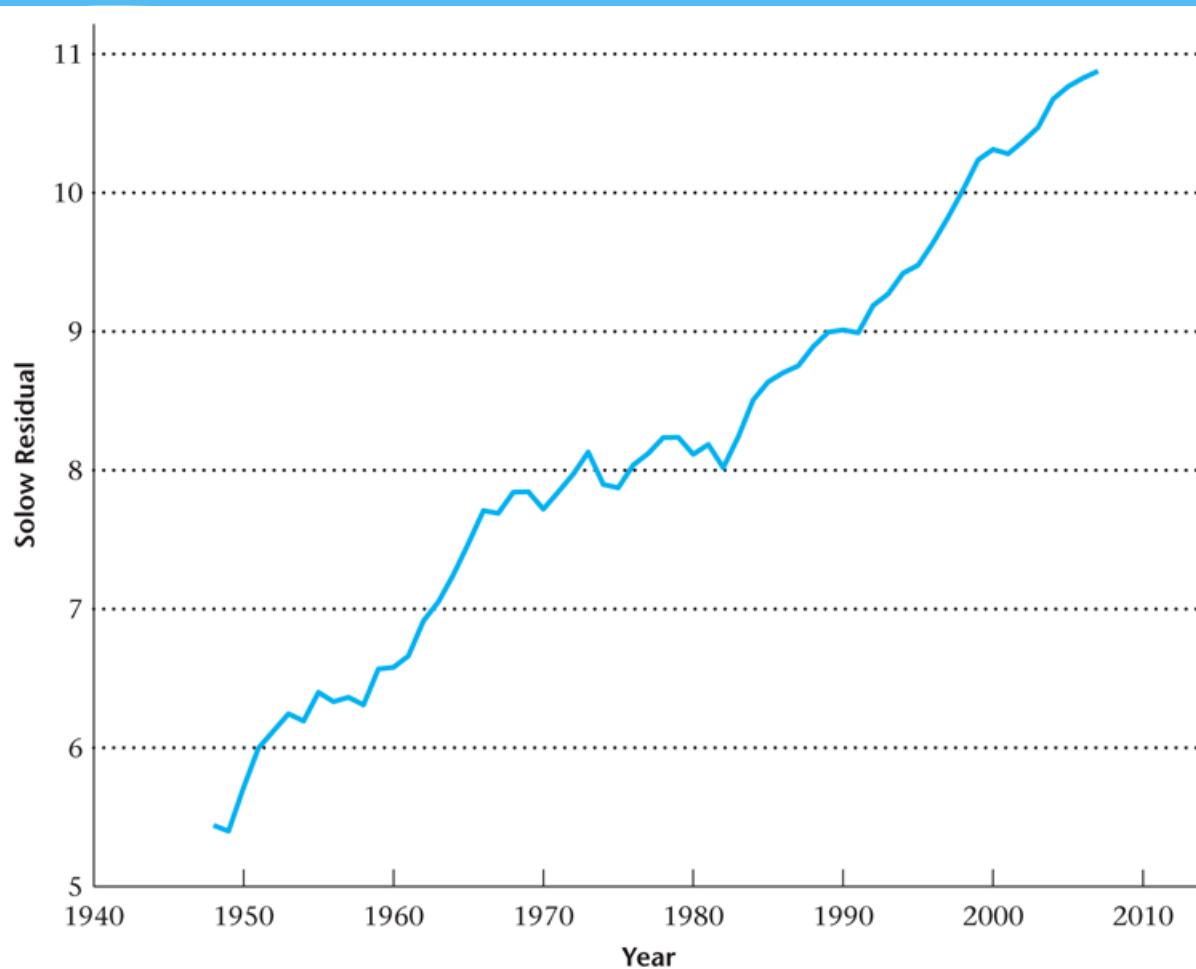
$$Y = zK^{0.36}N^{0.64}$$

Y, K and N are observed. We can recover z (the *Solow residual*):

$$z = Y / (K^{0.36}N^{0.64})$$

Figure 4.18

The Solow Residual for the United States



The shape of the production function

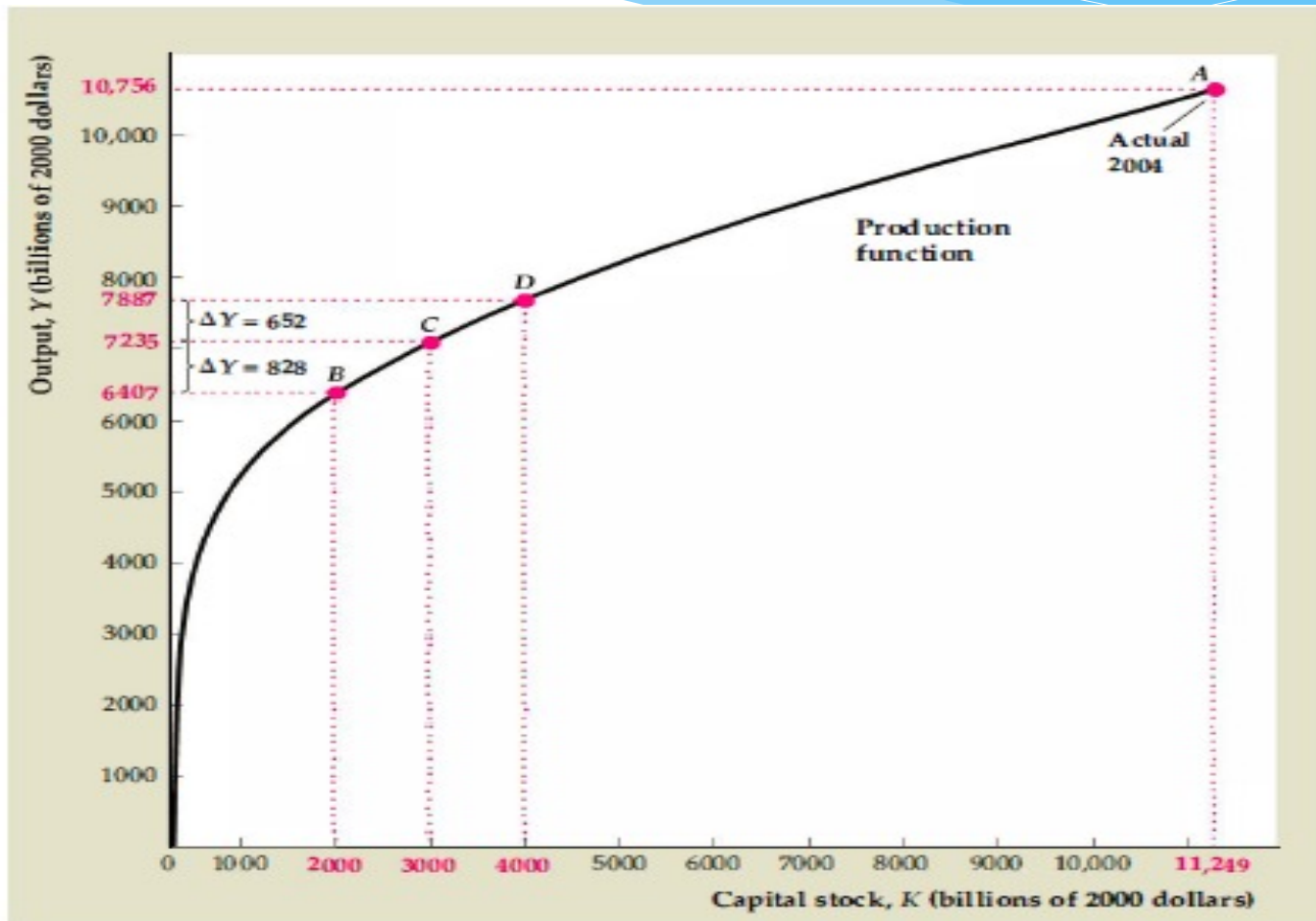
In 2004:

- $N = 139.3$ million workers
- $z = 20.68$

The Production function becomes:

$$\begin{aligned} Y &= zK^{0.3}N^{0.7} \\ &= 20.68K^{0.3}139.3^{0.7} \\ &= 655.11K^{0.3} \end{aligned}$$

The shape of the production function



Production function: output and capital

Two properties:

- output is increasing in capital
- the slope of the production function becomes flatter as capital increases

The marginal product of capital

Definition: the *marginal product of capital* is the additional output that can be produced with one additional unit of capital, holding labor constant

- It is the slope of the production function in the previous figure: output as a function of capital when labor is fixed

The marginal product of capital

Two properties of the production function can be expressed as:

1. $MPK > 0$.
2. MPK declines as the capital stock is increased: diminishing marginal productivity of capital.

Production function: output and labor

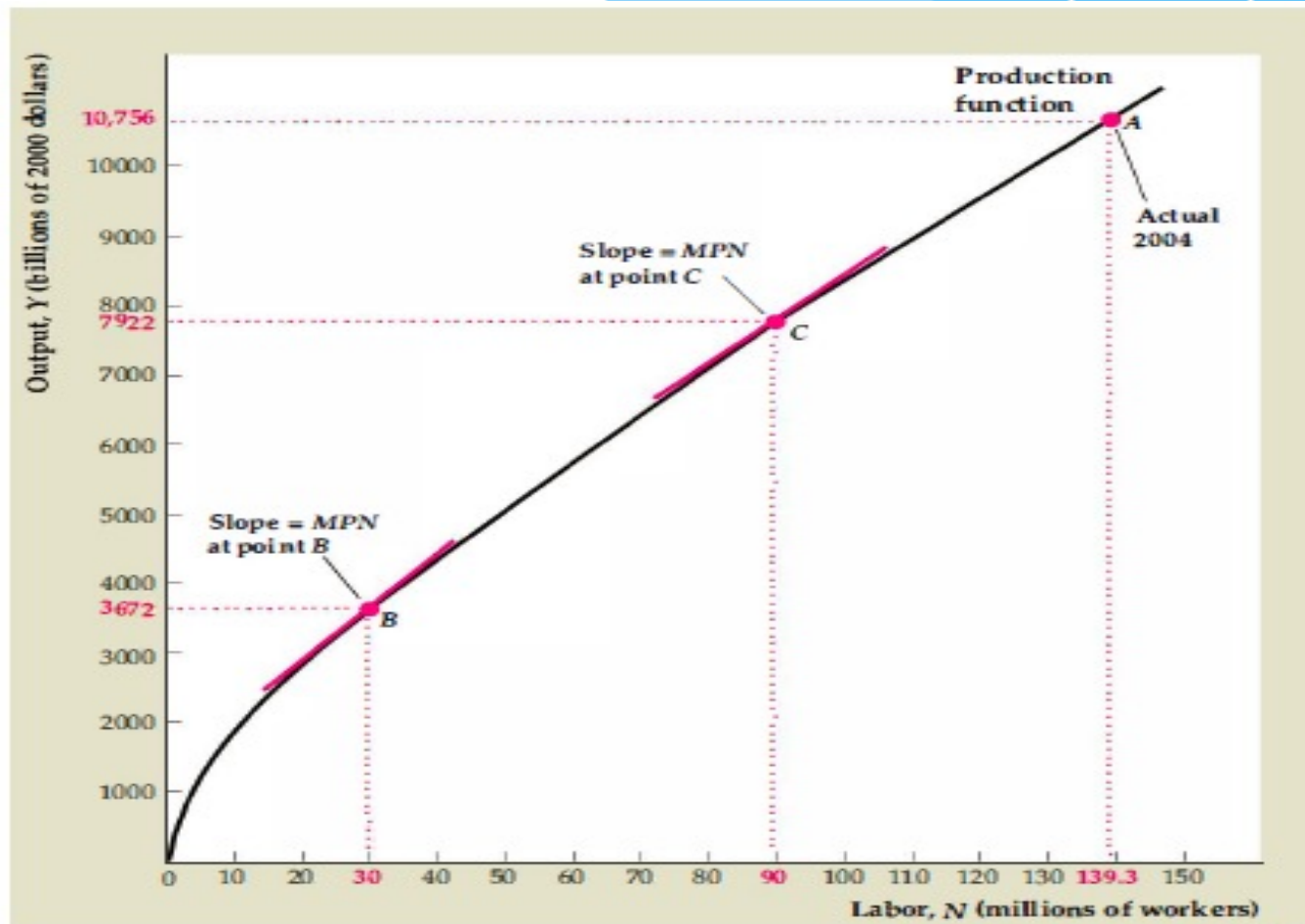
In 2004:

- $K = 11.249$ billion
- $z = 20.68$

The Production function becomes:

$$\begin{aligned} Y &= zK^{0.3}N^{0.7} \\ &= 20.68 (11.249^{0.3}) N^{0.7} \\ &= 339.54N^{0.3} \end{aligned}$$

The marginal product of labor



The marginal product of labor

Definition: the *marginal product of labor* is the additional output that can be produced with one additional unit of labor, holding capital constant

- It is the slope of the production function in the previous figure: output as a function of labor when capital is fixed.

Two properties

1. $MPL > 0$
2. MPL declines as the labor is increased:
diminishing marginal productivity of labor.

Profit maximization problem

- Firm rents capital at rate r .
- Firm hires labor at wage w .
 - wages of the workers they hired is determined in a competitive labor market and not set by the firms themselves
- Profits: Output minus Costs
 - a firm's goal is to earn the highest possible level of profit

$$\pi = zF(K, N) - rK - wN$$

- Assume that K is fixed: static economy

Profit maximization: example

- The Clip Joint is a small business that groom dogs:
 - capital: clippers, tubs, brushes
 - labor: workers that groomed dogs
 - output: number of groomed dogs

Profit maximization: example

# Workers N	# dogs groomed Y	Marginal Productivity of Labor MP_N
0	0	
1	11	11
2	20	9
3	27	7
4	32	5
5	35	3

Profit maximization: example

# Workers N	Output in \$ $P \times Y$	Marginal Productivity of Labor in \$ $P \times MP_N$
0	0	
1	330	330
2	600	270
3	810	210
4	960	150
5	1050	90

Profit maximization: example

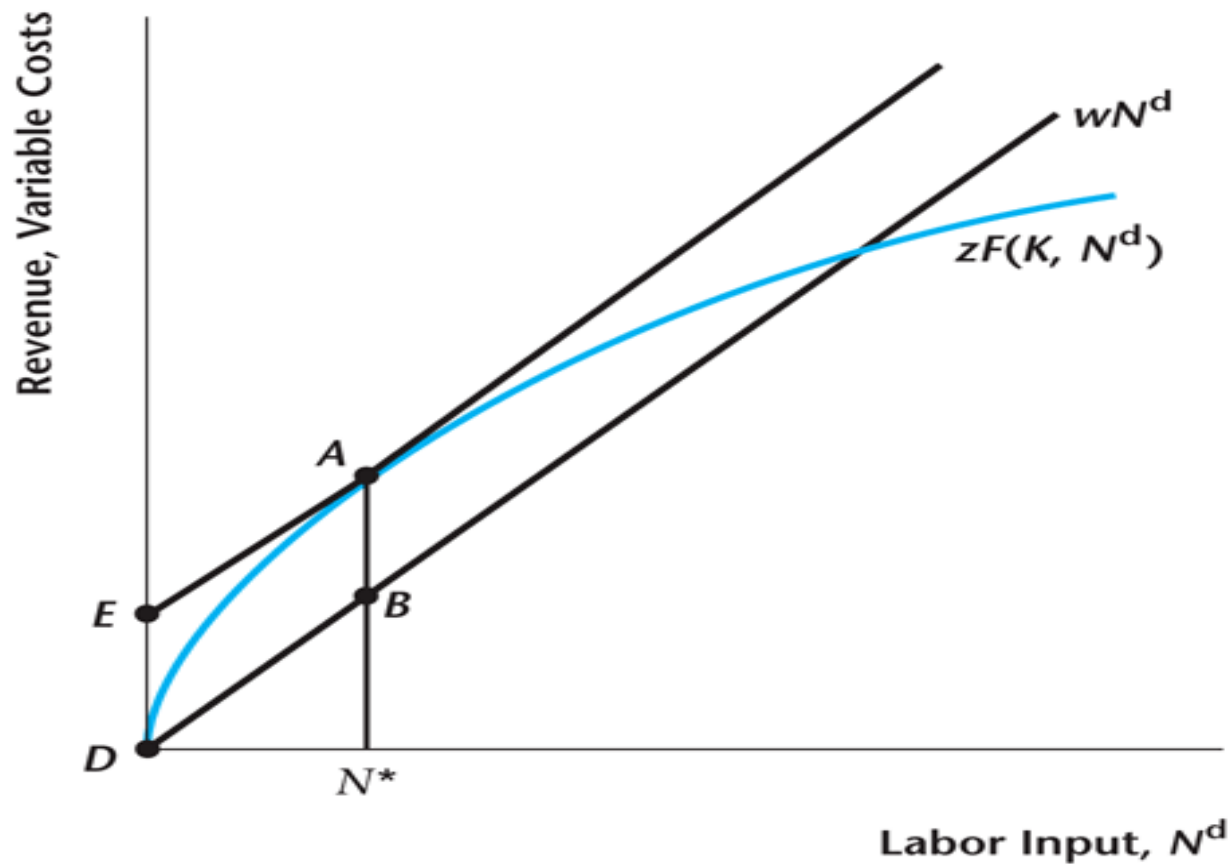
- $P = \$30$
- nominal wage: $w = \$240$
- real wage: $w / p = 240 / 30 = 8$ grooming per day
- $N^d = 2$

Profit maximization: example

- Impact of a decrease in the nominal wage $w = \$180$?
- $N^d = 3$.
- a decrease in the real wage rate raises labor demanded.

Figure 4.19

Revenue, Variable Costs, and Profit Maximization



Profit Maximization

When the firm maximizes profits, the marginal product of labor equals the real wage.

$$MPL = W$$

Figure 4.20

The Marginal Product of Labor Curve Is the Labor Demand Curve of the Profit-Maximizing Firm.

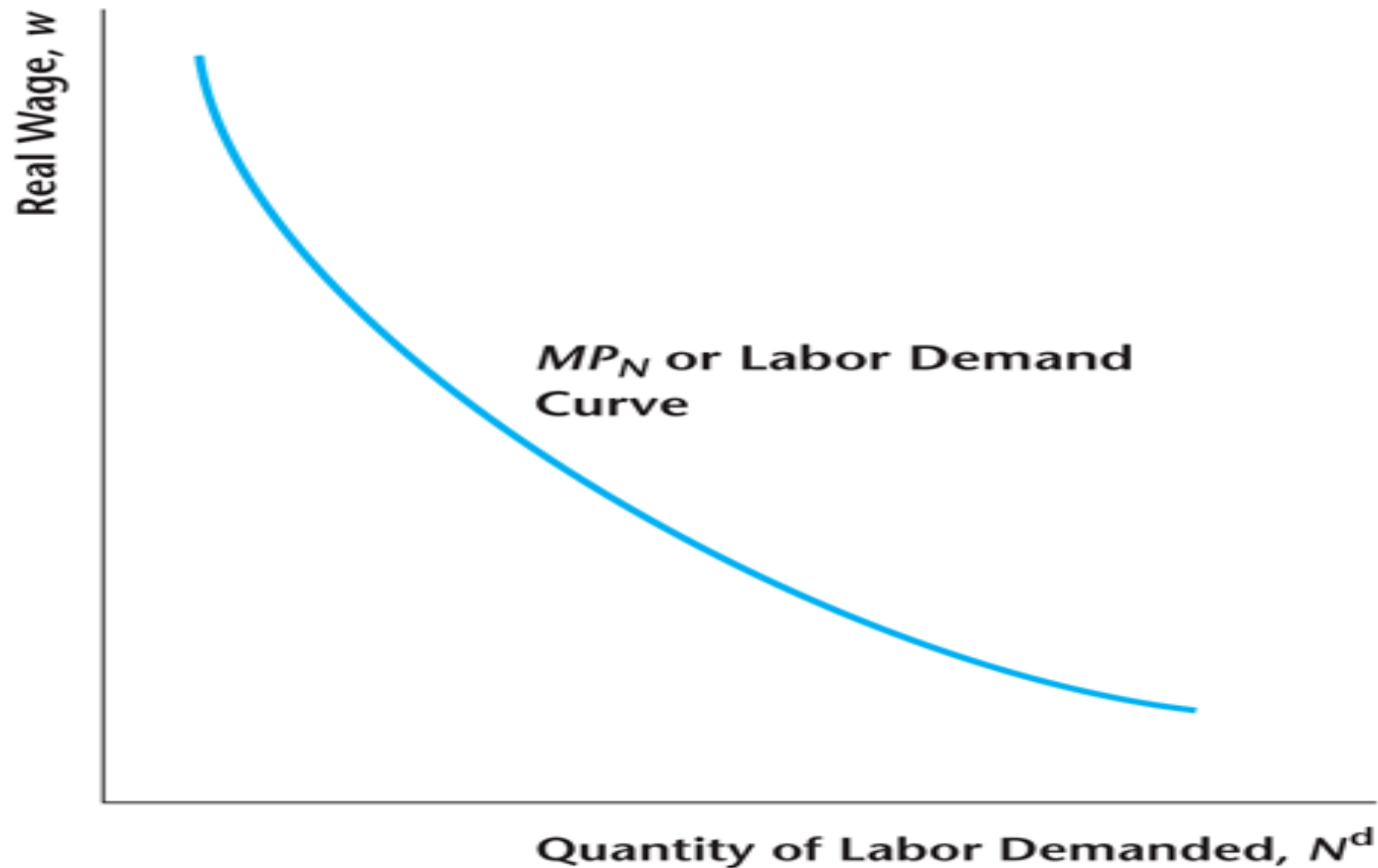


Figure 4.15

Adding Capital Increases the Marginal Product of Labor

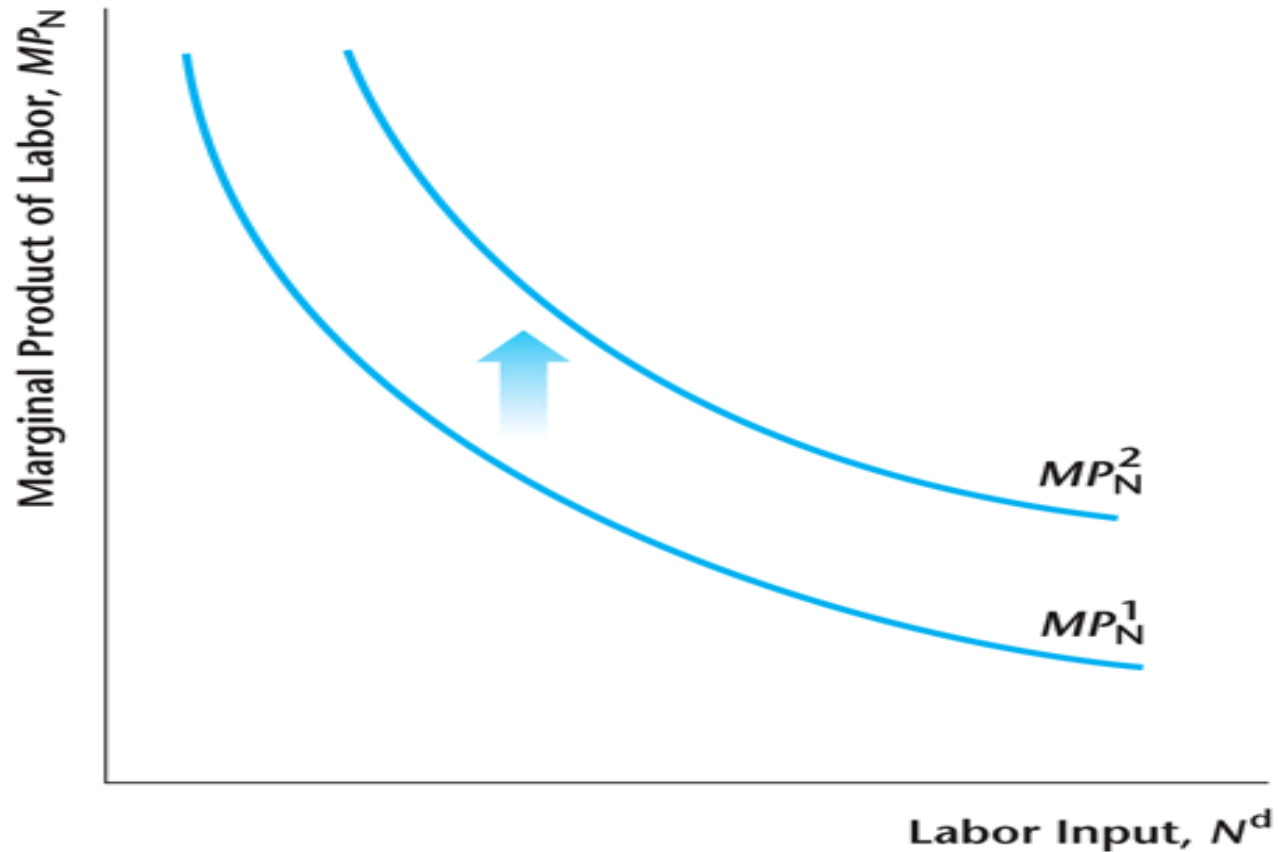


Figure 4.16
Total Factor Productivity Increases

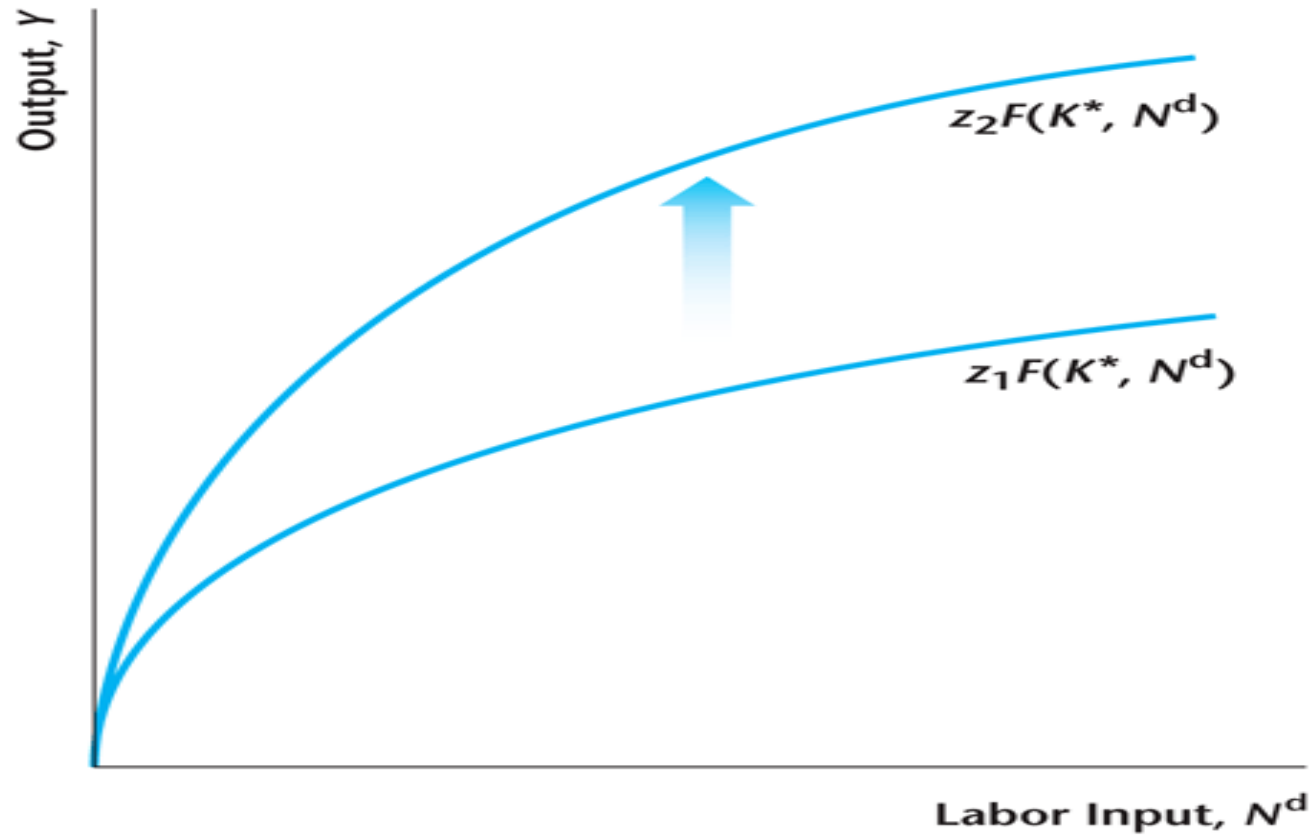
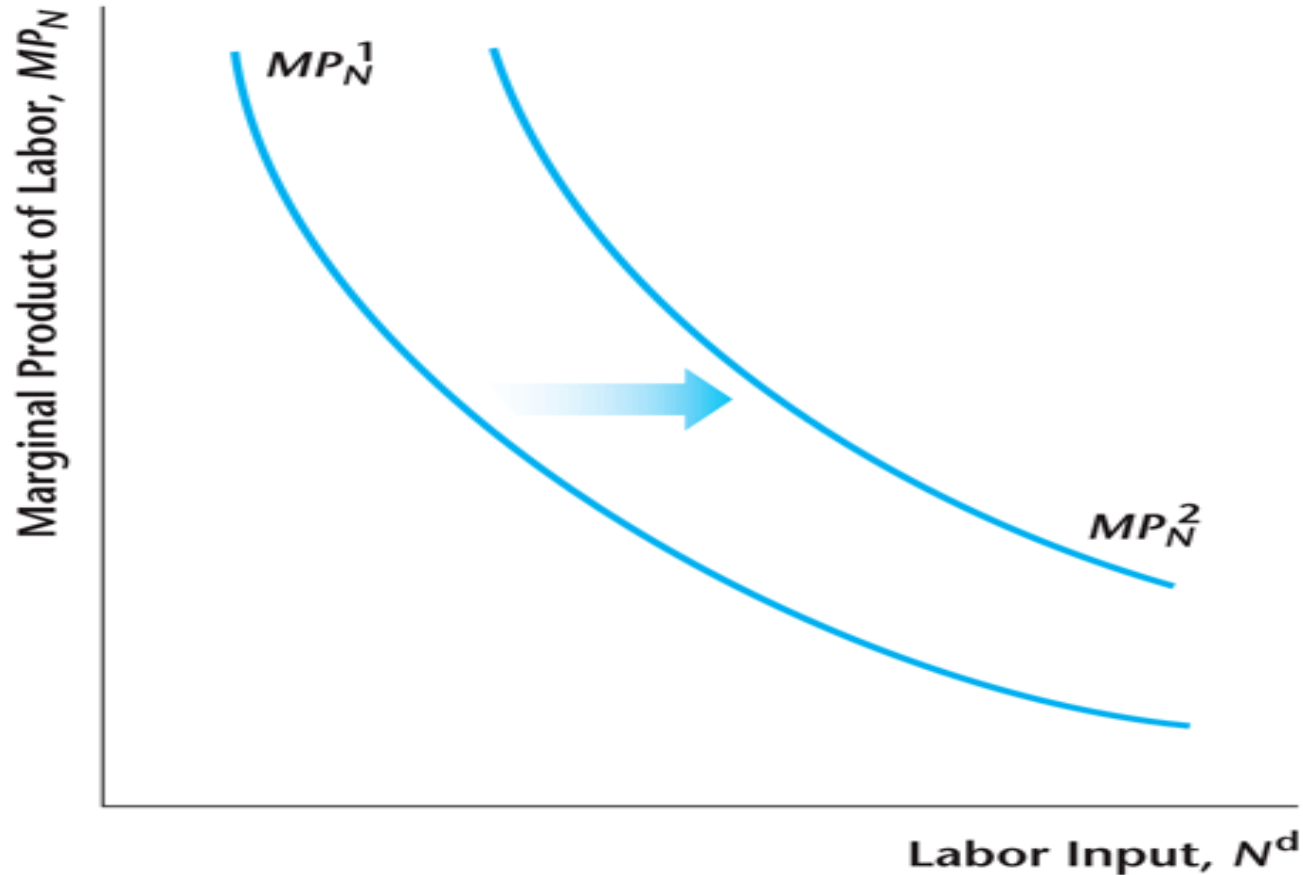


Figure 4.17

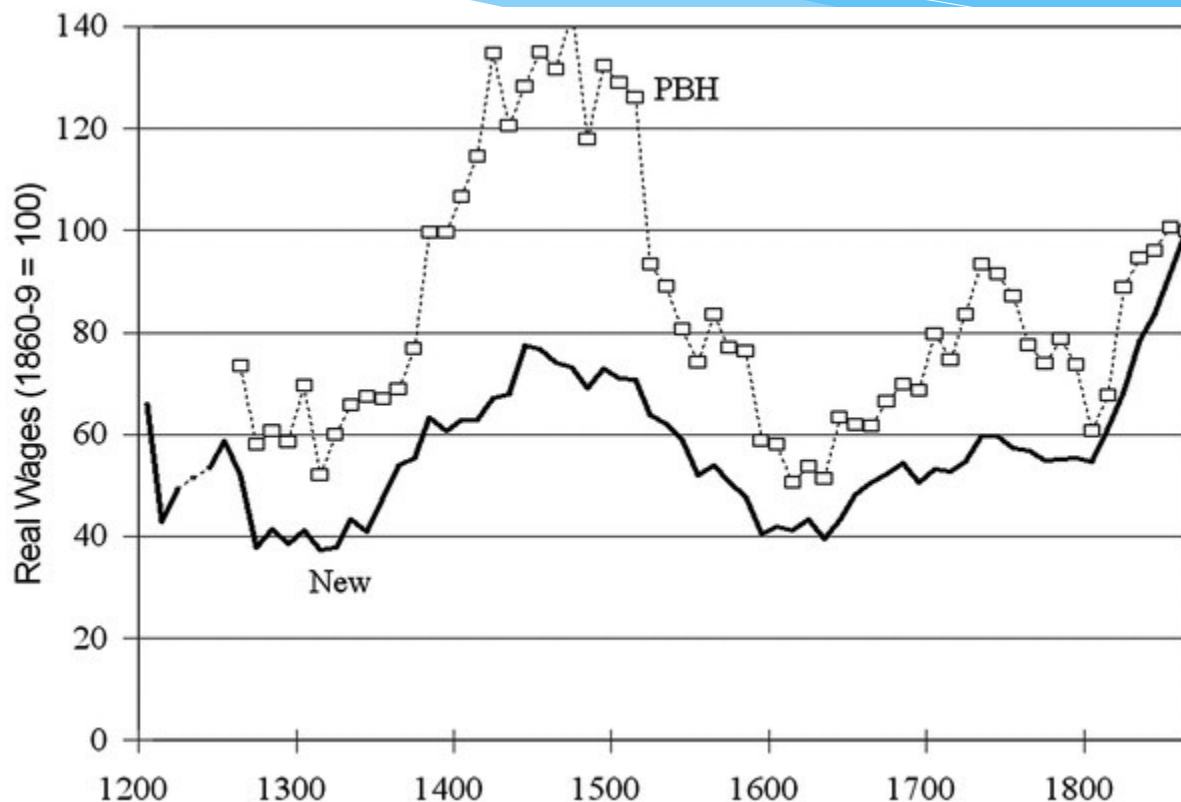
Effect of an Increase in Total Factor Productivity on the Marginal Product of Labor



Case study: The Black Death and Factor Prices

- Black Death: peaking in Europe between 1348 and 1350. It killed 30–60 percent of Europe's population.
- L falls, but K stays the same.
 - Fewer workers means that each remaining worker is more productive: MPL rises.
- Each machine becomes less productive: MPK falls.

Case study: The Black Death and Factor Prices



Source: Clark (2005), Journal of Political Economy. Phelps Brown and Hopkins vs. new series. In both series, 1860–69 has been set to 100.

Productivity in the 2008-09 Recession

- Problem: Timely measures of total factor productivity are not available, as we measure the capital stock with a lag.
- Can get timely measures of average labor productivity – closely related to total factor productivity, but not the same thing.

Puzzle

- In the 2008-09 recession, average labor productivity has declined much less than in typical recessions of the same severity.
- Why? Potential reasons are:
 1. The causes of the 2008-09 recession are different – housing sector problems and problems in the financial system.
 2. Long term shifts in employment across sectors – from construction and manufacturing to services.

Figure 4.21

Average Labor Productivity

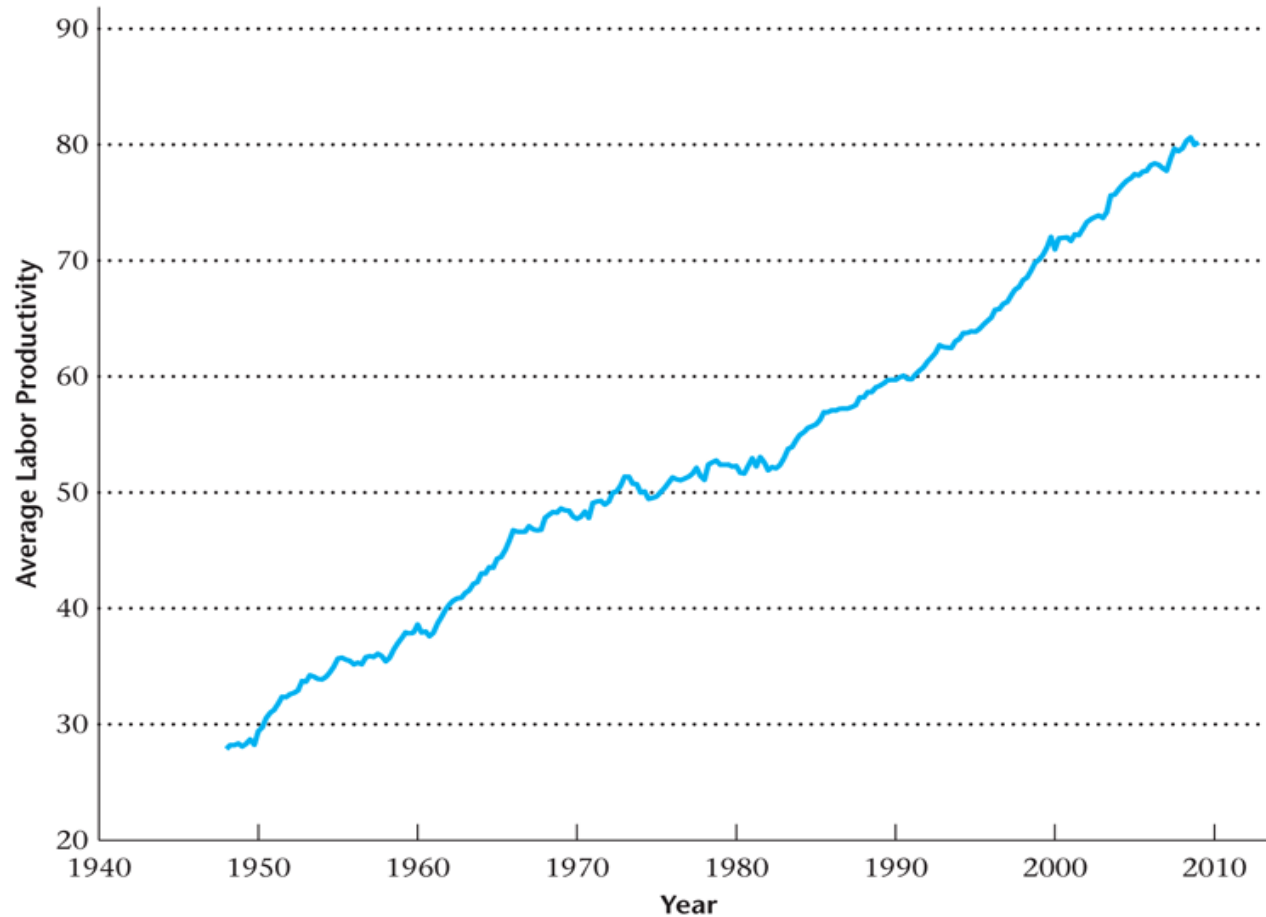
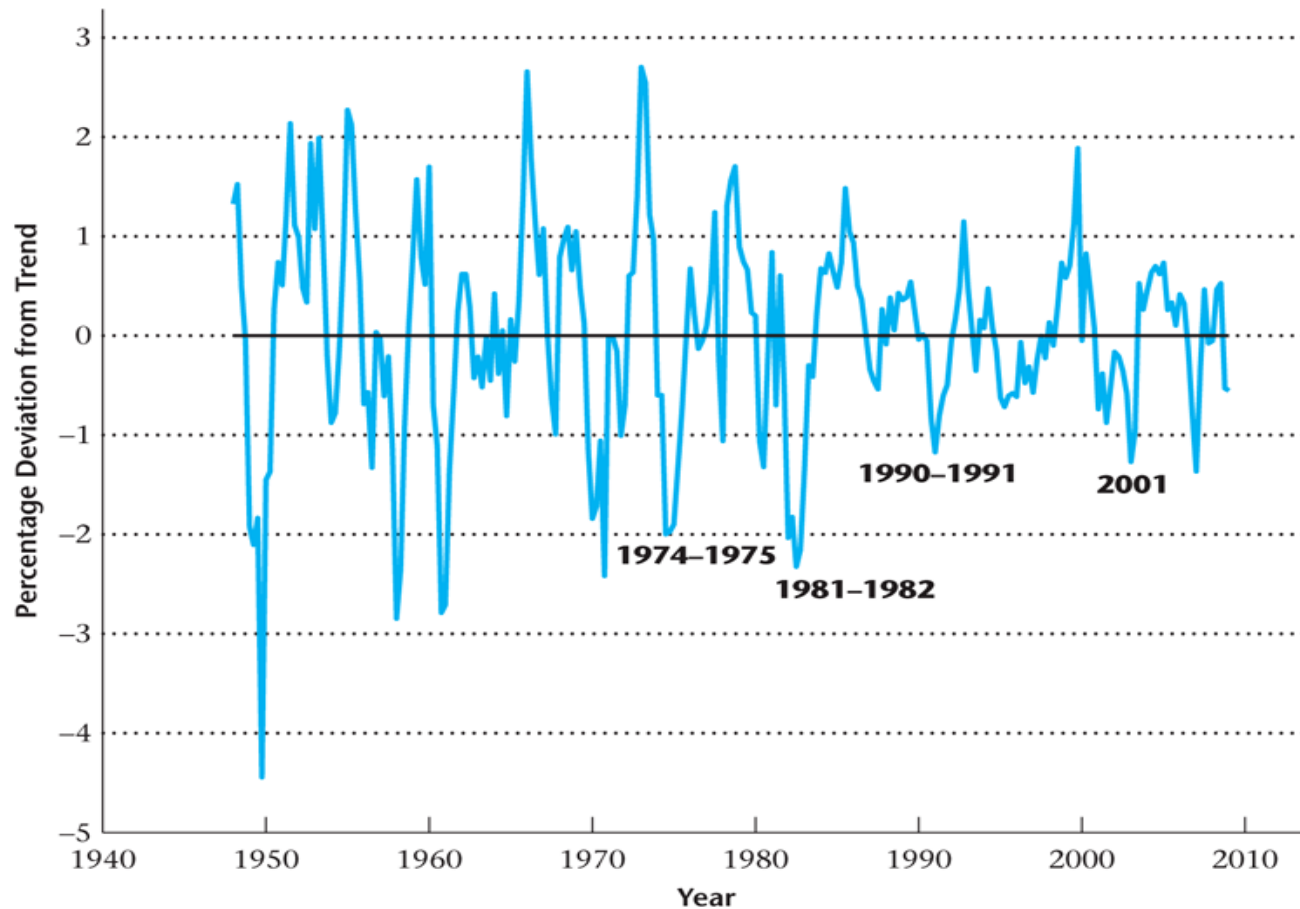


Figure 4.22

Percentage Deviations from Trend in Average Labor Productivity



Practice Question

“On Wednesday, Jeb Bush, the G.O.P. establishment’s standard-bearer, announced, as the centerpiece of his 2016 campaign, a plan to cut federal income-tax rates across the board.” New Yorker, Sep. 10.2015

Consider the partial equilibrium model of labor supply. The consumer has preferences over leisure l and consumption goods c represented by a utility function $U(c, l)$. The utility function satisfies regular conditions. The consumer has a dividend income $\pi > 0$. Suppose that the government imposes a proportional income tax on the representative consumer’s wage income. That is, the consumer’s wage income is $w(1 - t)(h - l)$ where t is the tax rate, w is the wage.

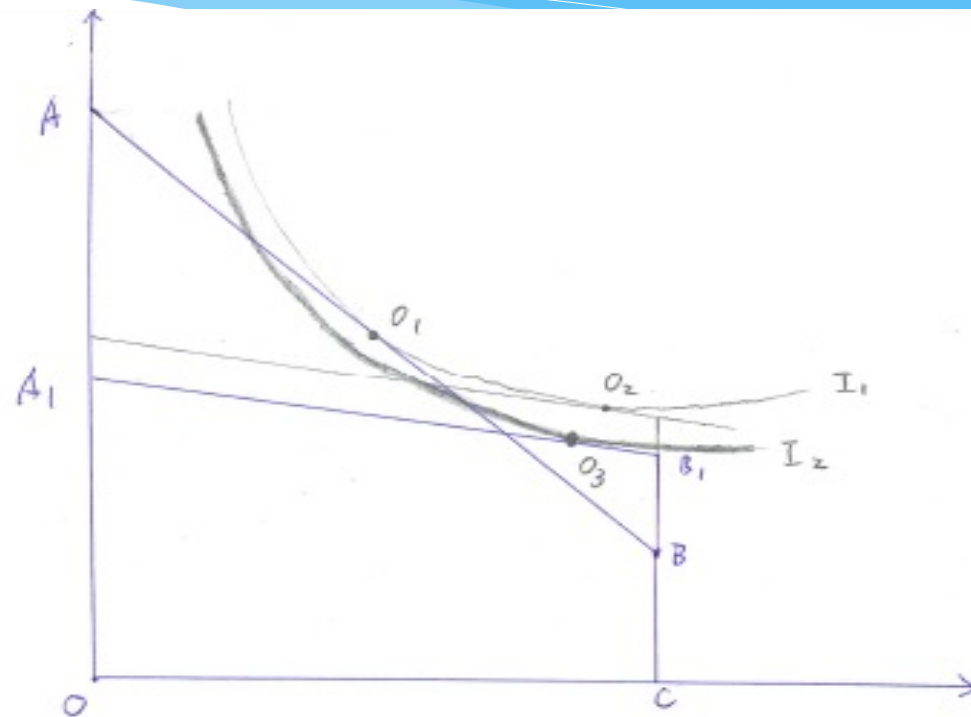
1. What is the budget constraint of workers? Represent it graphically.
2. Describe the impact of Bush’s plan on consumption and labor supply.

Practice Question

Consider the model of labor supply of Chapter 4. Suppose that the representative consumer's dividend income increases, and his or her wage rate falls at the same time.

Determine the effects on consumption and labor supply and explain your results in terms of income and substitution effects. (Assume the dividend income is always *greater* than the amount of Tax)

Practice Question

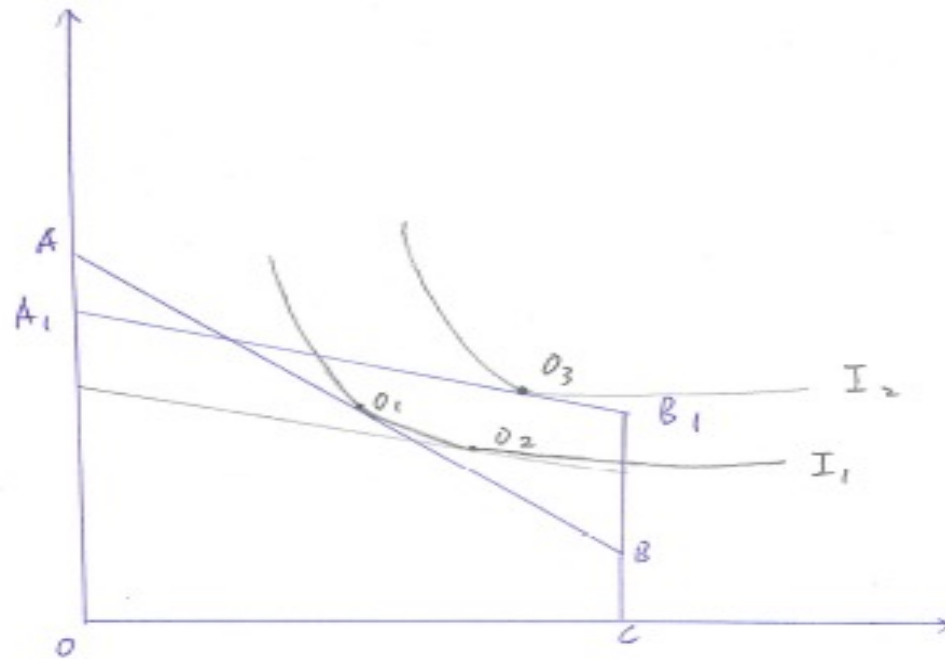


$O_1 \rightarrow O_2$ Substitution effect

$O_2 \rightarrow O_3$ income effect

$ABC \rightarrow A_1B_1C$

Practice Question



$O_1 \rightarrow O_2$ Substitution effect

$O_2 \rightarrow O_3$ Income effect.

$ABC - A_1B_1C$